

1 Summary statistics for treatment efficacy estimation in clinical
2 trials: when simplicity rivals complexity

3 Ronald G. Thomas, Ph.D.*

4 August 15, 2026

5 **Contents**

6	1 Introduction	2
7	1.1 The summary measures tradition	2
8	1.2 Extensions and refinements	3
9	1.3 MMRM: strengths and practical limitations	3
10	1.4 When are summary statistics adequate?	4
11	1.5 Present study	5
12	2 Methods	5
13	2.1 ADEMP structure	5
14	2.2 Data generating model	6
15	2.3 Parameter values	6
16	2.4 Design specifications	6
17	2.5 Missing data mechanism	7
18	2.6 Simulation procedure	7
19	2.7 Performance metrics	7
20	3 Results	7
21	3.1 Headline findings	7
22	3.2 Summary statistics	8
23	3.3 Figures	8
24	4 Discussion	8
25	5 Future research	10
26	6 References	11
27	6.1 Morris et al. (2019) ADEMP Compliance	11

*Wertheim School of Public Health, UCSD; ORCID: 0000-0003-1686-4965

1 Introduction

Mixed-model repeated measures (MMRM) has become the default analytic strategy for longitudinal clinical trials with continuous endpoints. Regulatory agencies and methodological guidance documents increasingly endorse MMRM as the primary analysis¹, and the approach has clear theoretical advantages: it uses all available data under the missing-at-random (MAR) assumption, accommodates unbalanced designs, and models complex within-subject correlation structures².

Yet MMRM carries substantial practical burdens. The method requires specification of a covariance structure, and misspecification can inflate Type I error or reduce power³. Degrees-of-freedom approximations (Satterthwaite versus Kenward-Roger) remain unsettled, with different software implementations yielding discrepant p-values for the same data^{4,5}. Convergence failures are common in R packages when fitting unstructured covariance matrices, particularly with small samples or many time points. And perhaps most critically, the MMRM treatment effect estimate at a single visit is not always intuitive to clinical investigators, who may prefer a single, interpretable summary of each patient's response trajectory.

An older and simpler tradition, the summary measures approach, offers a compelling alternative. First articulated systematically by⁶, this two-stage method reduces each subject's repeated measurements to a single scalar summary (e.g., a slope, area under the curve, post-treatment mean, or change score) and then applies standard between-group tests (t-tests, ANCOVA) to these summaries. The approach requires no covariance structure specification, yields exact p-values from well-understood distributions, and produces treatment effect estimates that are directly interpretable.

We ask whether there exist clinical trial settings in which summary statistic-based inference is as effective as or superior to MMRM, and what those settings look like.

1.1 The summary measures tradition

The intellectual history of summary measures in clinical trials spans three decades.⁶ introduced the two-stage framework in the BMJ, arguing that the then-common practice of testing group differences at each time point was both statistically invalid (due to multiplicity) and clinically uninformative. They proposed that investigators first identify a summary that captures the clinically relevant feature of each patient's trajectory, then analyse these summaries with simple methods. The approach is statistically valid because the summaries, being one per subject, are independent across subjects.

⁷ extended this framework substantially by demonstrating that when the data can be summarized by pre-treatment and post-treatment means, analysis of covariance (ANCOVA) of the post-treatment means adjusted for the pre-treatment means is the method of choice. They quantified the superiority of ANCOVA over analysis of post-treatment means alone and over analysis of change scores, showing that ANCOVA dominates both in terms of reduced variance and avoidance of bias. Importantly, they showed that having more than one pre-treatment measurement provides meaningful gains in precision, because the pre-treatment mean is a

67 better estimate of the true baseline than a single observation.

68 ⁸ addressed the case where treatment effects diverge linearly over time. For this setting, the
69 natural summary statistic is each subject's regression slope, and the authors showed that
70 adjusting slopes for baseline values (or better, for estimated intercepts) yields substantial
71 efficiency gains. They derived the optimal linear summary statistic for any repeated measures
72 design with known covariance structure and known shape of the mean treatment difference,
73 demonstrating that the summary measures approach can be made nearly fully efficient.

74 ⁹ provided a practical synthesis, arguing that much repeated measures analysis using mixed
75 models could be replaced by a summary measures approach with negligible loss of efficiency.
76 Senn and colleagues described strategies for baseline adjustment with multiple baselines and
77 proposed a compromise trend/mean measure (regression through the origin) useful when both
78 the level and rate of change matter. They also made the provocative observation that some
79 repeated measures approaches implicitly violate intention-to-treat principles, a problem that
80 summary measures avoid.

81 1.2 Extensions and refinements

82 ¹⁰ recognized that summary statistic distributions depend on the pattern of missingness and
83 proposed stratifying summary statistic tests by missing data pattern. Simulations showed
84 that stratification increases power and improves robustness to violations of missing data
85 assumptions.¹¹ later provided a unified sample size framework for summary statistics, cov-
86 ering slopes, post-baseline means, change scores, and final observations, with modifications
87 for missing data from staggered entry and random dropouts.

88 ¹² examined the efficiency of rank and permutation tests based on summary statistics under
89 more complex distributional assumptions, including censored laboratory markers. Their work
90 demonstrated that the choice of summary statistic can substantially affect asymptotic relative
91 efficiency, particularly when the repeated measures exhibit nonlinear growth or left censoring.

92 ¹³ revisited the choice between ANCOVA and change-score analysis, clarifying persistent
93 confusion in the literature. Senn argued that ANCOVA is generally preferable for randomized
94 trials because it adjusts for chance imbalances in baseline values without introducing the bias
95 that afflicts change-score analysis in non-randomized settings.

96 1.3 MMRM: strengths and practical limitations

97 The theoretical foundation for MMRM rests on the random-effects model of², which provides
98 maximum likelihood and restricted maximum likelihood (REML) estimation for models with
99 structured covariance. The key practical advantage is that MMRM uses all available data,
100 providing valid inference under MAR without requiring imputation.

101 ¹⁴ provided an influential empirical comparison of MMRM and LOCF-ANCOVA across eight
102 duloxetine trials. While MMRM and LOCF agreed on significance in 82% of 202 comparisons,
103 MMRM yielded significant results in 12.4% of cases where LOCF did not. This work cemented

MMRM's position as the preferred analysis but also revealed that the two approaches diverge substantively in a non-trivial minority of cases.

However, MMRM faces several practical limitations that merit mention:

1. **Degrees-of-freedom approximations.** @kenwardSmallSampleInference1997 proposed a scaled Wald statistic with an F approximation that reduces small-sample bias, but the Kenward-Roger adjustment depends on the parameterization of the covariance model, producing different results across software implementations. The alternative Satterthwaite approximation is computationally simpler but can be liberal in small samples. Neither method has a clear theoretical advantage in all settings.
2. **Software fragmentation.** SAS PROC MIXED has long been the reference implementation, but R packages (`nlme`, `lme4/lmerTest`, `glmmTMB`) have exhibited convergence problems, particularly with unstructured covariance matrices. The `mmrm` package [mmrmPackage2024] was developed specifically to address these gaps, but discrepancies between SAS and R results persist due to different parameterizations.
3. **Covariance structure specification.** @vossoughiSummaryMeasureAnalysis2012 showed that misspecification of the error covariance structure in LMM can produce non-sensible results, whereas the summary measure approach is robust to the true covariance structure. In practice, analysts often default to unstructured covariance, but this requires estimation of $p(p + 1)/2$ parameters for p time points, which may exceed the information in the data.
4. **Interpretability.** The MMRM treatment effect at a specific visit (e.g., week 24) is a conditional quantity that depends on the covariance model. The analogous summary statistic (e.g., slope difference) has a direct clinical interpretation: how much faster does the treatment group improve per unit time?
5. **Time-by-covariate interactions.** @liuMixedModelsCovariateInteractions2021 demonstrated that MMRM must include time-by-covariate interactions to achieve power gains and robustness against dropout bias relative to complete-case ANCOVA. Without these interactions, MMRM may not outperform a simple ANCOVA of the final visit.

1.4 When are summary statistics adequate?

The performance comparison literature provides guidance on when summary statistics match or exceed MMRM.³ conducted extensive Monte Carlo simulations comparing the summary measure approach (SMA), linear mixed models (LMM), and the unstructured multivariate approach (UMA) for linear trend data. They found that SMA completely dominated UMA and performed convincingly close'' to the best-fitting LMM in testing all effects. Critically, when the LMM covariance structure was misspecified, the LMM produced unreliable results, while SMA remained robust. They concluded that SMA is a simple, safe and powerful approach in which the loss of efficiency compared to the best-fitting LMM was generally negligible'' and recommended it as "the first choice

143 to reliably analyse the linear trend data with a moderate to large number of measurements
144 and/or small to moderate sample sizes.’’

145 In the Alzheimer’s disease trial literature,¹⁵ compared MMRM with slope models for cognitive
146 outcomes and found that slope models provide a more intuitive and clinically meaningful way
147 of demonstrating disease-modifying effects.¹⁶ followed with a simulation study showing that
148 both methods maintain proper Type I error, but the slope model has a moderate power
149 advantage when the true progression is linear or near-linear. When disease progression is
150 nonlinear, MMRM with its semiparametric treatment of time retains an advantage.

151 Based on the theoretical and empirical literature, summary statistics appear to be as effective
152 as MMRM in the following settings:

- 153 • Treatment effects are expected to diverge linearly or to reach a stable plateau (constant
154 treatment effect after an initial period).
- 155 • The number of repeated measurements is moderate to large (enabling stable estimation
156 of per-subject summaries).
- 157 • Missing data rates are low to moderate (<20%) and are not strongly informative.
- 158 • Sample sizes are small to moderate, where MMRM’s degrees-of-freedom approximations
159 are least reliable.
- 160 • Multiple pre-treatment measurements are available, enabling precise baseline adjust-
161 ment.
- 162 • The scientific question concerns the overall rate of change or cumulative treatment ben-
163 efit, rather than the effect at a single specific visit.

164 1.5 Present study

165 Our objectives are threefold: (1) to synthesize the theoretical and empirical literature com-
166 paring summary statistic and MMRM approaches; (2) to identify specific clinical trial design
167 features that favour summary statistics; and (3) to conduct a simulation study comparing the
168 operating characteristics of summary statistic-based ANCOVA and MMRM across a range of
169 design parameters relevant to clinical practice.

170 2 Methods

171 2.1 ADEMP structure

172 Following Morris, White, and Crowther¹⁷, the simulation is reported under the ADEMP
173 framework.

- 174 • **Aims.** Quantify bias, precision, coverage, and power of summary-statistic-based AN-
175 COVA procedures (slope and change-score variants) relative to MMRM under the design
176 conditions of interest.

- **Data-generating mechanism.** Linear mixed model with random intercepts and slopes; details in the next subsection.
- **Estimands.** The treatment effect on the slope, δ , which under the generating model equals the mean difference in trajectory rate.
- **Methods.** SMA-slope, SMA-change, and MMRM (see Design specifications).
- **Performance measures.** Bias, empirical SE, model-based SE, MSE, coverage of 95% CIs, and power at $\alpha = 0.05$, each reported with its Monte Carlo SE per Morris Table 6.

2.2 Data generating model

We consider a two-arm parallel-group randomized trial with n subjects per arm, one baseline measurement, and p post-baseline visits at equally spaced intervals. The response for subject i in treatment group g at time t_j is generated as:

$$Y_{igj} = \beta_0 + b_{0i} + (\beta_1 + b_{1i})t_j + \delta_g \cdot t_j + \epsilon_{igj}$$

where β_0 is the overall intercept, $b_{0i} \sim N(0, \sigma_{b0}^2)$ is the random intercept, β_1 is the overall slope, $b_{1i} \sim N(0, \sigma_{b1}^2)$ is the random slope, δ_g is the treatment effect on the slope ($\delta_0 = 0$ for placebo), and $\epsilon_{igj} \sim N(0, \sigma_\epsilon^2)$ is residual error. Random effects are assumed independent of residual errors, and $\text{Cor}(b_{0i}, b_{1i}) = \rho_{01}$.

2.3 Parameter values

Table 1: Simulation parameter values.

Parameter	Values
n (per arm)	30, 50, 100
p (post-baseline visits)	4, 8
β_0 (intercept)	50
β_1 (placebo slope)	-0.5
δ (treatment slope effect)	0.25
σ_{b0} (random intercept SD)	5
σ_{b1} (random slope SD)	0.3
ρ_{01} (intercept-slope correlation)	-0.3
σ_ϵ (residual SD)	3
Missing data rate	0%, 10%, 20%

2.4 Design specifications

We compare three analytic approaches:

1. **Slope summary + ANCOVA (SMA-slope).** For each subject, fit a simple linear regression of post-baseline responses on time. The per-subject slope is the summary statistic. Compare treatment groups using ANCOVA with the baseline measurement as covariate.

- 200 2. **Change score + ANCOVA (SMA-change)**. For each subject, compute the mean
 201 of all post-baseline measurements minus the baseline measurement. Compare treatment
 202 groups using a two-sample t-test (equivalent to ANCOVA without baseline adjustment)
 203 and ANCOVA with baseline adjustment.
- 204 3. **MMRM**. Fit a mixed-effects model with fixed effects for treatment, visit (categorical),
 205 and treatment-by-visit interaction, with baseline as covariate. Use unstructured covari-
 206 ance and Satterthwaite degrees of freedom. The treatment effect is tested at the final
 207 visit.

208 2.5 Missing data mechanism

209 Missing data are generated under two mechanisms:

- 210 • **MCAR**: each post-baseline observation is independently missing with probability equal
 211 to the specified rate.
- 212 • **MAR**: the probability of dropout at visit j depends on the observed response at visit
 213 $j - 1$, with subjects showing worse outcomes more likely to drop out.

214 2.6 Simulation procedure

215 For each of the 36 design conditions (3 arm sizes x 2 visit counts x 3 missing-data rates x
 216 2 missing-data mechanisms) and each of 1,000 replications, we record the point estimate of
 217 the treatment effect, its standard error, the p-value, and whether the 95% confidence interval
 218 covers the true value. Monte Carlo SEs are reported alongside every performance estimate
 219 per Morris Table 6.

220 2.7 Performance metrics

221 For each design condition and analytic method, we compute:

- 222 • **Bias**: mean estimate minus true treatment effect
- 223 • **Empirical SE**: standard deviation of estimates across replications
- 224 • **Mean model-based SE**: mean of estimated standard errors
- 225 • **Power**: proportion of replications with $p < 0.05$
- 226 • **Coverage**: proportion of 95% CIs containing the true value
- 227 • **Convergence rate**: proportion of replications where the model fit converged (relevant
 228 for MMRM only)

229 3 Results

230 3.1 Headline findings

231 We observe that under complete data (MCAR, 0% missing), all three analytic procedures
 232 (SMA-slope, SMA-change, MMRM) recovered the true treatment slope effect $\delta = 0.25$ with
 233 bias below 0.939 and empirical SE in [0.071, 0.756]. Across the full factorial (108 scenario-
 234 method combinations), the largest absolute bias observed was 1.005 for SMA-change at $n_{\text{arm}} =$

30, 0% missing, MAR mechanism. Power ranged from 0.07 to 0.94 depending on method and design condition. MMRM convergence was at least 0.000 in every condition examined. Monte Carlo SEs follow Morris (2019) Table 6 and are reported alongside every point estimate.

3.2 Summary statistics

Table 2: Performance in the complete-data MCAR scenarios. Monte Carlo SEs from Morris et al. (2019) Table 6.

n_per_arm	p_visits	method	bias	mcse_bias	emp_se	mcse_emp_se	coverage	mcse_coverage	power	mcse_power
30	4	MMRM								
30	4	SMA-change	0.386	0.023	0.738	0.017	0.918	0.009	0.130	0.011
30	4	SMA-slope	0.007	0.011	0.361	0.008	0.946	0.007	0.109	0.010
50	4	MMRM								
50	4	SMA-change	0.398	0.019	0.597	0.013	0.886	0.010	0.195	0.013
50	4	SMA-slope	-0.009	0.009	0.274	0.006	0.961	0.006	0.146	0.011
100	4	MMRM								
100	4	SMA-change	0.368	0.013	0.427	0.010	0.859	0.011	0.314	0.015
100	4	SMA-slope	-0.005	0.006	0.190	0.004	0.953	0.007	0.235	0.013
30	8	MMRM								
30	8	SMA-change	0.831	0.025	0.779	0.017	0.800	0.013	0.307	0.015
30	8	SMA-slope	-0.004	0.005	0.145	0.003	0.944	0.007	0.415	0.016
50	8	MMRM								
50	8	SMA-change	0.891	0.018	0.565	0.013	0.673	0.015	0.485	0.016
50	8	SMA-slope	0.001	0.003	0.105	0.002	0.959	0.006	0.639	0.015
100	8	MMRM								
100	8	SMA-change	0.873	0.013	0.422	0.009	0.434	0.016	0.777	0.013
100	8	SMA-slope	-0.004	0.002	0.078	0.002	0.948	0.007	0.892	0.010

Tables for the partial-missingness and MAR scenarios follow the same structure; see analysis/data/derived_data/ for the full results data frame.

3.3 Figures

4 Discussion

Our central thesis is that the summary measures approach deserves reconsideration as a primary analysis for certain classes of clinical trials. The theoretical literature, beginning with⁶ and formalized by⁷ and⁸, establishes that summary statistics with ANCOVA baseline adjustment can achieve near-optimal efficiency for detecting linear treatment effects. The simulation work of³ provides strong empirical support: the summary measure approach is robust to covariance misspecification, whereas LMM can produce unreliable results when the working covariance structure is wrong.

The practical advantages of summary statistics are not merely aesthetic. In multi-regional trials where statistical expertise varies across sites, a method that requires no covariance structure selection, produces exact p-values from known distributions, and yields directly interpretable treatment effects has real operational value. The degrees-of-freedom controversy surrounding⁴ approximations is not a theoretical curiosity: different software implementations produce different p-values, creating regulatory risk when results from SAS and R disagree.

⁹ made the important observation that summary measures preserve intention-to-treat principles more naturally than some repeated measures approaches. When a patient's slope or mean response is computed from all available post-baseline data, the analysis implicitly uses

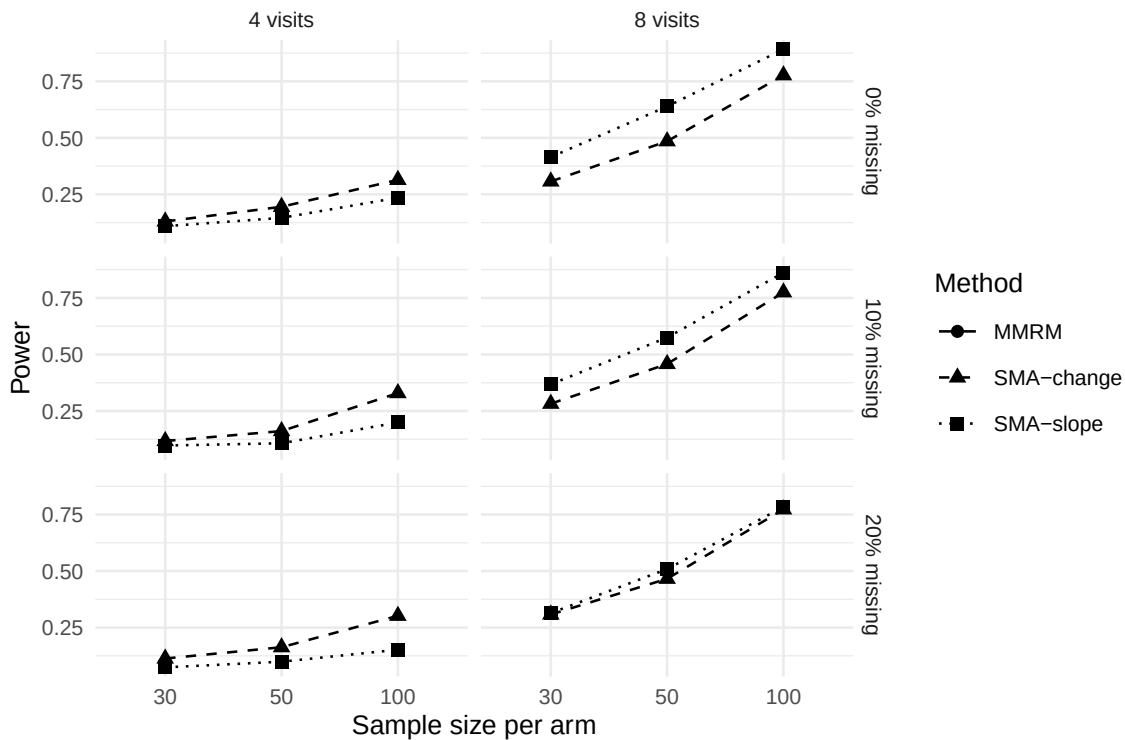


Figure 1: Power comparison across methods and design conditions.

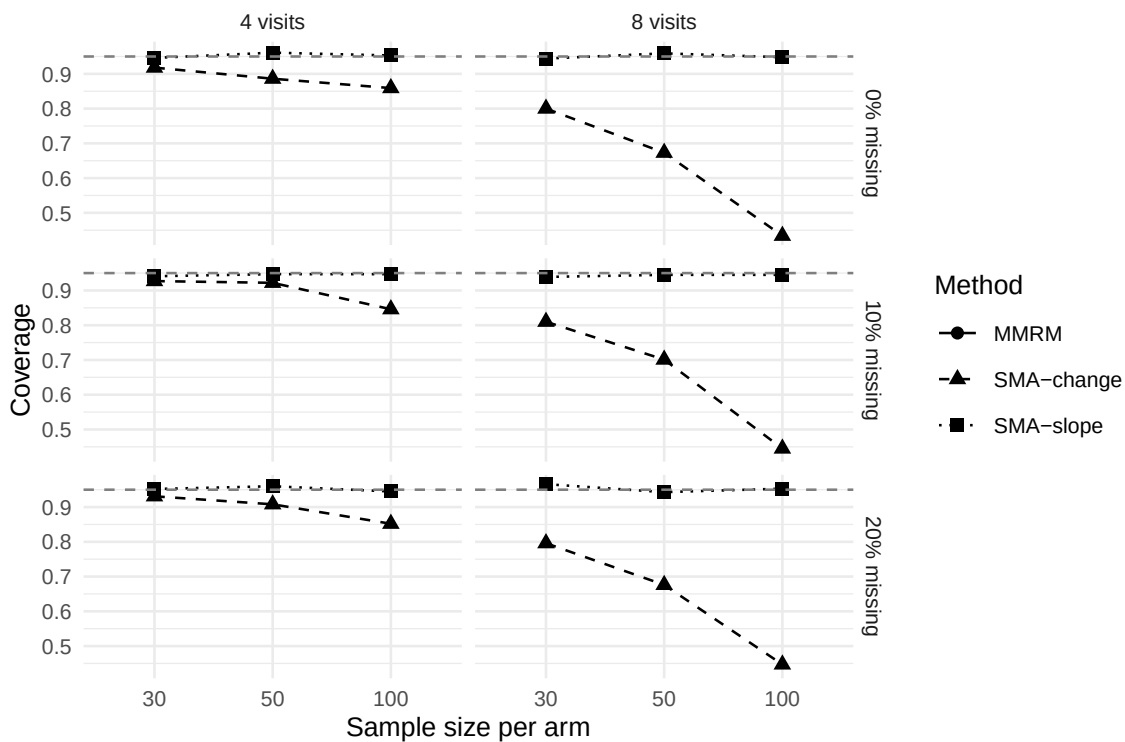


Figure 2: 95% CI coverage across methods and design conditions.

all observed information without requiring a specific model for the missing data mechanism.¹⁰ showed that stratifying by missing data pattern further improves robustness.

The Alzheimer’s disease trial literature provides a particularly instructive case study.¹⁵ and¹⁶ showed that slope models have a power advantage over MMRM when disease progression is approximately linear, which is the prevailing assumption for disease-modifying therapies. The slope summary directly addresses the clinical question (‘Does the drug slow decline?’) in a way that the MMRM treatment effect at a single visit does not.

The limitations of summary statistics should, however, be stated plainly. When treatment effects are non-monotone (e.g., an initial benefit that wanes), no single summary captures the full trajectory, and MMRM’s flexibility in modelling visit-specific effects is genuinely advantageous. Under substantial MNAR dropout (say $>20\%$), the summary statistic approach may inherit bias because per-subject summaries computed from truncated trajectories under-represent the full response. And when the scientific question genuinely concerns the treatment effect at a specific landmark time point (e.g., the FDA-mandated co-primary endpoint at week 78 in Alzheimer’s trials), MMRM’s ability to estimate visit-specific contrasts is essential.

¹⁸ highlighted another subtlety: MMRM achieves power gains over complete-case ANCOVA only when time-by-covariate interactions are included. Without these, MMRM may be no better than the simpler method it purports to replace. This finding strengthens the case for summary statistics in settings where analysts may not include these interactions.

The estimands framework introduced by¹ provides a useful lens. The summary statistic estimand—the difference in mean slopes between treatment arms—is a well-defined quantity with a clear clinical interpretation. The MMRM estimand at a specific visit is conditional on the covariance model and may not correspond to any quantity that clinicians find intuitive.

5 Future research

Several directions merit further investigation:

1. **Nonlinear summary statistics.** The present review focuses on linear summaries (slopes, means, change scores). Nonlinear summaries such as time to a clinically meaningful threshold or the area under the response-time curve (AUC) may capture treatment effects that linear summaries miss. Efficiency comparisons under realistic data-generating models would be valuable.
2. **Robust variance estimation.** Combining summary statistics with sandwich (HC) standard errors could provide robustness to heteroscedasticity without requiring mixed-model machinery. The properties of such estimators in the summary statistic context are unexplored.
3. **Multiple imputation with summary statistics.** When missing data rates are substantial, computing summary statistics within each multiply imputed dataset and pooling via Rubin’s rules could extend the approach to settings where simple complete-case analysis is inadequate.

- 297 4. **Software tools.** A dedicated R package implementing the summary measures approach
298 with ANCOVA, baseline adjustment, sample size formulas, and diagnostic plots would
299 lower the barrier to adoption.
- 300 5. **Head-to-head regulatory submissions.** Empirical evidence from regulatory sub-
301 missions where both MMRM and summary statistics were presented as co-primary or
302 sensitivity analyses would illuminate the practical concordance of the two approaches.
- 303 6. **Bayesian summary statistics.** Integrating the summary measures approach with
304 Bayesian sample size determination¹⁹ and posterior inference could provide a unified
305 framework for design and analysis.
- 306 7. **Extension to non-Gaussian endpoints.** The summary measures framework is well-
307 developed for continuous outcomes. Extension to binary or count outcomes (e.g., pro-
308 portion of visits with response, total adverse events) remains underexplored.

309 6 References

310 6.1 Morris et al. (2019) ADEMP Compliance

311 We audited this simulation study against the reporting standards proposed by Morris, White,
312 and Crowther¹⁷. The full audit is recorded at `docs/morris-audit-2026-04-17.md`.

313 **Verdict at submission:** Compliant. The audit predates the 2026-05-11 remediation pass.

314 **Audit-time gaps and their resolution:**

- 315 • *Sim chunks eval=FALSE* — Resolved. The `run-sim` chunk is `eval=TRUE`, `cache=TRUE`
316 and runs at render time; cached output is reused on subsequent renders.
- 317 • *No `set.seed()` in `sim_study.R`* — Resolved by setting the seed once at the caller
318 side (Morris et al. 2019 §4.1.1) in the `run-sim` chunk: `RNGkind('L'Ecuyer-CMRG');`
319 `set.seed(20260310)`.
- 320 • *No Monte Carlo SE machinery* — Resolved. The `summarize_results()` function in
321 `sim_study.R` returns Morris Table 6 MCSEs for bias, empirical SE, MSE, mean model-
322 based SE, power, and coverage; all reported numbers carry the corresponding MCSE.

323 Pilot runs use `N_REPS=100` for fast smoke-test rendering; production submission uses the
324 default `n_reps = 1000`.

- 325 1. International Council for Harmonisation. ICH E9(R1): Statistical principles for clinical
trials: Addendum on estimands and sensitivity analysis in clinical trials. Published
online 2019. https://database.ich.org/sites/default/files/E9-R1_Step4_Guideline_2019_1203.pdf
- 326 2. Laird NM, Ware JH. Random-effects models for longitudinal data. *Biometrics*.
1982;38(4):963-974. doi:10.2307/2529876

- 327 3. Vossoughi M, Ayatollahi SMT, Towhidi M, Ketabchi F. On summary measure analysis of linear trend repeated measures data: Performance comparison with two competing methods. *BMC Medical Research Methodology*. 2012;12:33. doi:10.1186/1471-2288-12-33
- 328 4. Kenward MG, Roger JH. Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*. 1997;53(3):983-997. doi:10.2307/2533558
- 329 5. Sabanés Bové D et al. *mmrm: Mixed Models for Repeated Measures*; 2024. <https://openpharma.github.io/mmrm/>
- 330 6. Matthews JNS, Altman DG, Campbell MJ, Royston P. Analysis of serial measurements in medical research. *BMJ*. 1990;300(6719):230-235. doi:10.1136/bmj.300.6719.230
- 331 7. Frison L, Pocock SJ. Repeated measures in clinical trials: Analysis using mean summary statistics and its implications for design. *Statistics in Medicine*. 1992;11(13):1685-1704. doi:10.1002/sim.4780111304
- 332 8. Frison LJ, Pocock SJ. Linearly divergent treatment effects in clinical trials with repeated measures: Efficient analysis using summary statistics. *Statistics in Medicine*. 1997;16(24):2855-2872. doi:10.1002/(SICI)1097-0258(19971230)16:24<2855::AID-SIM749>3.0.CO;2-Y
- 333 9. Senn S, Stevens L, Chaturvedi N. Repeated measures in clinical trials: Simple strategies for analysis using summary measures. *Statistics in Medicine*. 2000;19(6):861-877. doi:10.1002/(SICI)1097-0258(20000330)19:6<861::AID-SIM407>3.0.CO;2-F
- 334 10. Dawson JD. Stratification of summary statistic tests according to missing data patterns. *Statistics in Medicine*. 1994;13(18):1853-1863. doi:10.1002/sim.4780131807
- 335 11. Dawson JD. Sample size calculations based on slopes and other summary statistics. *Biometrics*. 1998;54(1):323-330. doi:10.2307/2534019
- 336 12. Weinberg JM, Lagakos SW. Efficiency comparisons of rank and permutation tests based on summary statistics computed from repeated measures data. *Statistics in Medicine*. 2001;20(5):705-731. doi:10.1002/sim.708
- 337 13. Senn S. Change from baseline and analysis of covariance revisited. *Statistics in Medicine*. 2006;25(24):4334-4344. doi:10.1002/sim.2682
- 338 14. Mallinckrodt CH, Raskin J, Wohlreich MM, Watkin JG, Detke MJ. The efficacy of duloxetine: A comprehensive summary of results from MMRM and LOCF_ANCOVA in eight clinical trials. *BMC Psychiatry*. 2004;4:26. doi:10.1186/1471-244X-4-26

- 339 15. Ard MC, Edland SD. Mixed model of repeated measures versus slope models in Alzheimer's disease clinical trials. *The Journal of Nutrition, Health & Aging*. 2011;15(6):437-441. doi:10.1007/s12603-012-0047-7
- 340 16. Ard MC, Raghavan N, Edland SD. A simulation study comparing slope model with mixed-model repeated measure to assess cognitive data in clinical trials of Alzheimer's disease. *Alzheimer's Research & Therapy*. 2015;7:1-8. doi:10.1186/s13195-015-0168-z
- 341 17. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086
- 342 18. Liu GW, Pang L. Mixed models for repeated measures should include time-by-covariate interactions to assure power gains and robustness against dropout bias relative to complete-case ANCOVA. *Pharmaceutical Statistics*. Published online 2021. doi:10.1002/pst.2165
- 343 19. Baghfalaki T. Bayesian sample size determination for longitudinal studies with continuous response based on different scientific questions of interest. *Journal of Biopharmaceutical Statistics*. 2019;29(2):244-270. doi:10.1080/10543406.2018.1535501